



Information Technology Institute (ITI)
Analog IC Design

Lab 06

Differential Amplifier

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1 Part 1: Differential Amplifier Design

1.1 Design Specifications

The specs:

Parameter	Value
Supply (V_{DD})	1.8 V
Current seed (ref. current)	20 μ A
Bias current (I_{SS})	40 μ A (20 μ A per side)
Differential gain	8 V/V
CM output level	$V_{DD}/3 = 0.6$ V
Load capacitance	1 pF

Table 1: Differential amplifier design specifications

Since the output level (0.6 V) is close to ground, I used a **PMOS input pair**.

Choosing R_D

$V_{OUT-CM} = I_D \cdot R_D$, and since $I_D = 20\mu A$

$$R_D = \frac{V_{OUT-CM}}{I_D} = \frac{0.6 \text{ V}}{20\mu A} = \boxed{30 \text{ k}\Omega} \quad (1)$$

Setting r_o and Finding g_m/I_D

We will choose L to set $r_o \gg R_D \rightarrow r_o = 10 \times R_D$. $|A_v| \approx g_m(R_D || r_o)$ so $|A_v| \approx 0.91 g_m R_D$. will be $g_m R_D = (g_m/I_D) \cdot I_D \cdot R_D = (g_m/I_D) \cdot V_{RD}$ where $V_{RD} = I_D R_D = 0.6V$

$$\frac{g_m}{I_D} = \frac{|A_v|}{0.91 \times V_{RD}} = \frac{8}{0.91 \times 0.6} = \boxed{14.65 \text{ V}^{-1}} \quad (2)$$

1.2 Input Pair Sizing via Sizing Assistant

V_{DS} of the tail current source to 300 mV to allow more output swing. So $I_D = 20\mu A$, $g_m/I_D = 8/(0.91 \times 0.6) = 14.65$, $r_o = 10 \times (0.6/I_D)$, $V_{DS} = V_{DD} - V_{DS,tail} - V_{OUT-CM} = 1.8 - 0.3 - 0.6 = 0.9$ V, and $V_{SB} = 0.3$ V = VDD - (VDD-0.3)

```
[01-08-2026 18:06:24] - Info: [Sizing Assistant] knob deviation (GM/ID) = 2.07% Condition deviation (RO) = 4.37%
[01-08-2026 18:06:45] - Info: [Sizing Assistant] The resultant point 'L = 350n'
[01-08-2026 18:06:49] - Info: [Sizing Assistant] The resultant point 'W = 30u'
[01-08-2026 18:07:01] - Info: [Sizing Assistant] The resultant point 'VTH = 870.2m'
[01-08-2026 18:07:06] - Info: [Sizing Assistant] The resultant point 'VGS = 941.1m'
[01-08-2026 18:07:13] - Info: [Sizing Assistant] The resultant point 'VSTAR = 140.5m'
[01-08-2026 18:07:20] - Info: [Sizing Assistant] The resultant point 'GM = 284.8u'
[01-08-2026 18:07:43] - Info: [Sizing Assistant] The resultant point 'VDSAT = 122.5m'
```

Figure 1: Sizing Assistant log for the input pair

Parameter	Value
L (LP2)	350 nm
W (WP2)	30 μm
V_{TH}	870.2 mV
V_{GS}	941.1 mV
$V_{STAR} = V_{ov}$	140.5 mV
G_M	284.8 $\mu\text{A}/\text{V}$
V_{DSAT}	122.5 mV

Table 2: Input pair sizing result from SA

1.3 Common-Mode Input Level

calculate the required CM input level. $V_{s,cm} = V_{DD} - V_{DS,tail} = 1.8 - 0.3 = 1.5$ V. The input pair source-gate voltage is $V_{SG} = V_{GS} = 941.1$ mV So:

$$V_{ICM} = V_{s,cm} - V_{SG} = 1.5 - 0.9411 = \boxed{0.559 \text{ V}} \quad (3)$$

1.4 Tail Current Source Design

current = $20\mu\text{A}$, mismatch $\sigma(I_{out})/I_{out} \leq 2\%$, $\lambda = 0.1$, compliance voltage ≤ 200 mV, minimize area. as the current mirror lab. $\sigma(I_{out})/I_{out}$.

The mirror output transistor (M2, carrying the $40\mu\text{A}$ tail current) has *twice* the area of the reference transistor (M3, carrying $20\mu\text{A}$), so $m = 2$ in the mismatch formula:

$$\frac{\sigma(I_{out})}{I_{out}} = \sqrt{1 + \frac{1}{m} \cdot \frac{idmis}{I_D}} \times 100 = \sqrt{1.5} \cdot \frac{idmis}{I_D} \times 100 \quad (4)$$

The L , W , $AREA$ and λ vs g_m/I_D for both the 1% and 2% mismatch curves: $g_m/I_D = 10$ since V_{comp} is 200mV

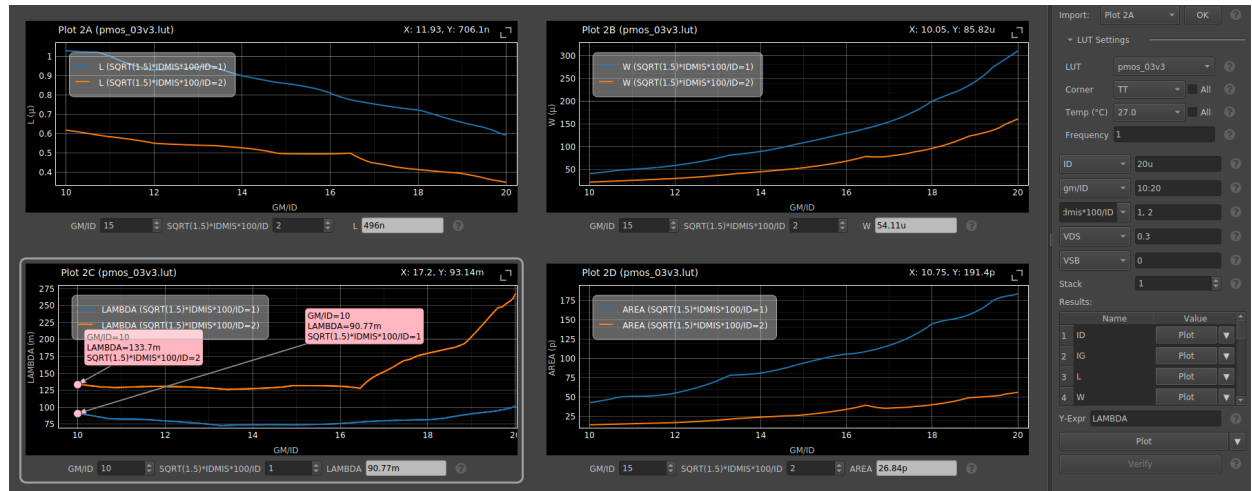


Figure 2: SA sweep: L , W , λ , $Area$ vs g_m/I_D for 1% and 2% mismatch curves, cursor at $g_m/I_D = 10$

as we see the λ is bigger than 0.1 when the mismatch=2 so we will pick mismatch 1.

Curve	λ @ $g_m/I_D = 10$	Meets $\lambda \leq 0.1$?	Decision
1% mismatch	90.77 m	Yes ($0.0908 < 0.1$)	Picked
2% mismatch	133.7 m	No ($0.1337 > 0.1$)	Rejected

Table 3: Mismatch curve comparison at the compliance-limited $g_m/I_D = 10$

1.4.1 tail sizing

Plot 2A Plot 2B Plot 2C Plot 2D

Import: Plot 2A OK ?

▼ LUT Settings

LUT pmos_03v3 ?

Corner TT All ?

Temp (°C) 27.0 All ?

Frequency 1 ?

ID 20u ?

gm/ID 10 ?

sqrt(1.5)*idr 1 ?

VDS 0.3 ?

VSB 0 ?

Stack 1 ?

Results:

	Name	TT-27.0
1	ID	20u
2	IG	N/A
3	L	1.03u
4	W	41.28u
5	VDB	300m
6	VGB	942.3m
7	VGS	942.3m
8	VDS	300m

Y-Expr LAMBDA ?

Plot ▼

Verify ?

Figure 3: Final SA settings/result for the tail mirror on the 1% mismatch curve at $g_m/I_D = 10$

Parameter	Value
L (reference & mirror)	$1.03 \mu\text{m}$
W (reference, WP1)	$41.28 \mu\text{m}$
W (mirror output, $2\times\text{WP1}$)	$82.56 \mu\text{m}$
V_{GS}	942.3 mV
V_{DS}	300 mV
λ	90.77 m

Table 4: Final tail current source sizing

1.5 CMIR Hand Calculation

Calculate the min and max CM input levels. Is the previously selected CM input level in the valid range?

A: $V_{DSAT,tail} = 0.1527\text{V}$ and $V_{DSAT,pair} = 119.2\text{mV}$:

- $V_{ICM,max} \approx V_{DD} - V_{DSAT,tail} - V_{GS,in} = 1.8 - 0.1527 - 0.941 = 0.706 \text{ V}$
- $V_{ICM,min} \approx V_{OUT-CM} + V_{DSAT,in} - V_{GS,in} = 0.6 + 0.1192 - 0.941 = -0.235 \text{ V}$

2.1 Testbench Schematic

The screenshot displays a SPICE simulation of a differential amplifier circuit. The circuit includes a central **DIFFPAIR** block with inputs **VINN** and **VINP**, and outputs **VOU TP** and **VOU TN**. The circuit is powered by **VDD** (1.8V) and **AVDD** (1.8V). The input differential mode is **VIDIFF** (0V), and the common-mode input is **VICM** (0.559V). The output differential mode is **VODIFF** (0V), and the common-mode output is **VOCM** (0.5861V). The simulation results show the output voltages **VOUTP** and **VOUTN** are approximately 0.5861V.

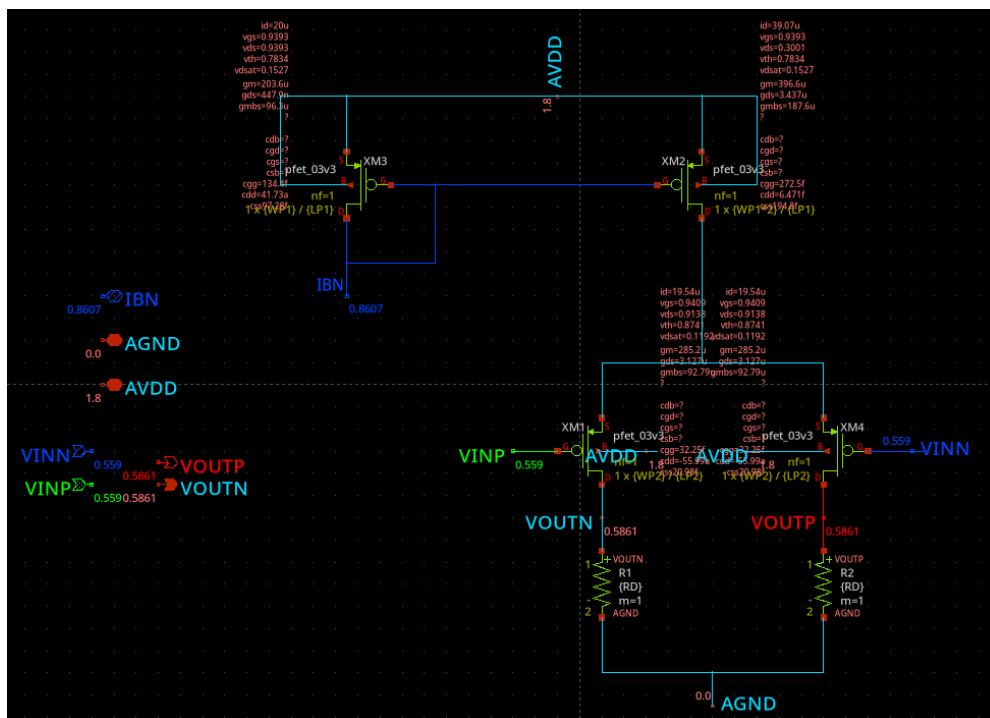
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.include $::180MCU_MODELS/design.spice
.lib $::180MCU_MODELS/sm141064.spice typical

```

OP

2.2 OP Simulation



ITI

Param	M1 (input)	M4 (input)	M2 (mirror out)	M3 (ref, diode)
I_D	19.54 μA	19.54 μA	39.07 μA	20.00 μA
V_{GS}	940.9 mV	940.9 mV	939.3 mV	939.3 mV
V_{DS}	913.8 mV	913.8 mV	300.1 mV	939.3 mV
V_{TH}	874.1 mV	874.1 mV	783.4 mV	783.4 mV
V_{DSAT}	119.2 mV	119.2 mV	152.7 mV	152.7 mV
$V^* (=2/gmID)$	137.0 mV	137.0 mV	197.0 mV	196.5 mV
g_m/I_D	14.60	14.60	10.15	10.18
G_M	285.2 $\mu\text{A/V}$	285.2 $\mu\text{A/V}$	396.6 $\mu\text{A/V}$	203.6 $\mu\text{A/V}$
G_{DS}	3.127 $\mu\text{A/V}$	3.127 $\mu\text{A/V}$	3.437 $\mu\text{A/V}$	447.9 nA/V
G_{MB}	92.79 $\mu\text{A/V}$	92.79 $\mu\text{A/V}$	187.6 $\mu\text{A/V}$	96.3 $\mu\text{A/V}$
Region	All 4 transistors: $V_{DS} > V_{DSAT} \Rightarrow$ SATURATION ✓			

Table 5: OP simulation results for all four transistors

2.3 Differential Small-Signal AC Analysis

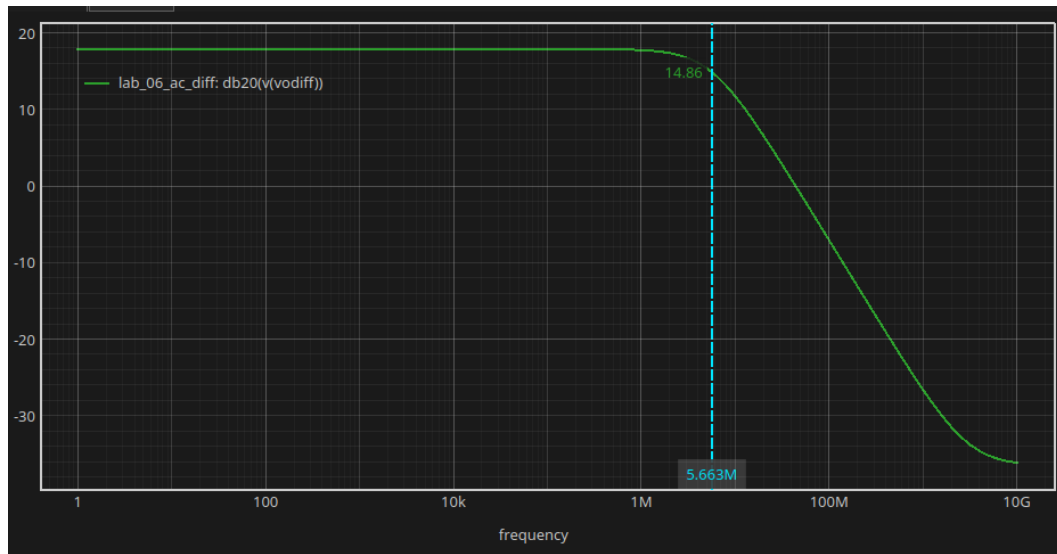


Figure 6: Differential AC gain Bode plot (dB), -3dB point marked at 5.663 MHz

$r_o = 1/g_{ds} = 1/3.127\mu = 319.8 \text{ k}\Omega$, so $R_D || r_o = 30\text{k} || 319.8\text{k} = 27.63 \text{ k}\Omega$.

$$|A_{v,diff}|_{hand} = g_m(R_D || r_o) = 285.2\mu \times 27.63\text{k} = 7.88 \text{ V/V} = 17.93 \text{ dB} \quad (5)$$

$$f_{-3dB,hand} = \frac{1}{2\pi(R_D || r_o)C_L} = \frac{1}{2\pi \times 27.63\text{k} \times 1\text{p}} = 5.76 \text{ MHz} \quad (6)$$

Quantity	Hand Analysis	Simulation
DC diff gain (V/V)	7.88	7.82
DC diff gain (dB)	17.93	17.865
Bandwidth (MHz)	5.76	5.684

Table 6: Differential gain & BW: hand analysis vs simulation

2.4 Common-Mode Small-Signal AC Analysis

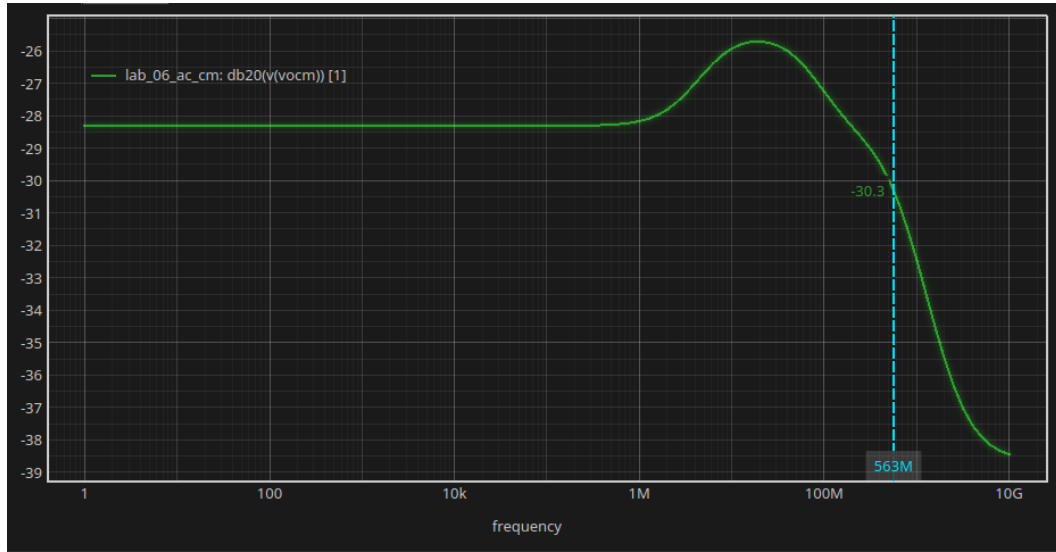


Figure 7: CM AC gain Bode plot (dB) – flat at low freq, peaks around tens of MHz, then rolls off

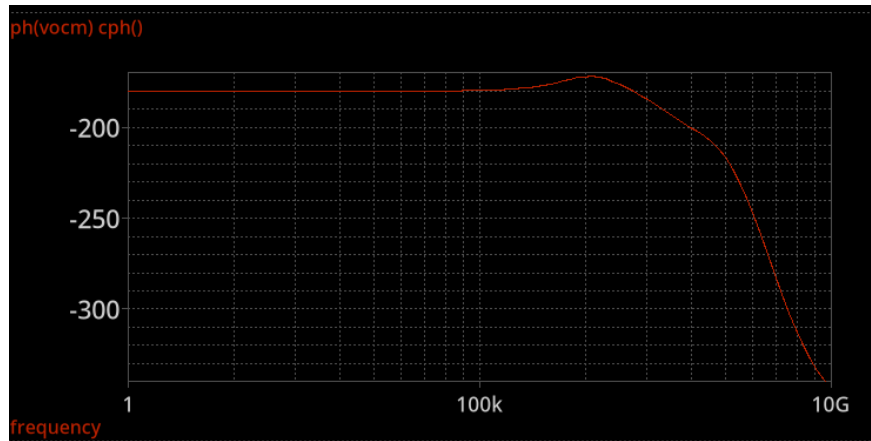


Figure 8: CM AC phase response

$$R_{ss} = r_{o,tail} = 1/g_{ds,M2} = 1/3.437\mu = 290.9 \text{ k}\Omega:$$

$$A_{vcm,hand} \approx \frac{-R_D}{2R_{ss} + 1/g_{m1}} = \frac{-30k}{2(290.9k) + 1/285.2\mu} = \frac{-30k}{585.3k} = -0.0513 \text{ V/V} = -25.8 \text{ dB} \quad (7)$$

- $R_{ss} = r_{o2} = 1/g_{ds2}$
- C_{tail} is primarily the sum of the tail transistor's drain capacitance and the input pair's source capacitances: $C_{tail} \approx c_{dd2} + c_{ss1} + c_{ss4}$

The impedance expression is:

$$Z_{ss} = R_{ss} \parallel \frac{1}{sC_{tail}} = \frac{R_{ss}}{1 + sR_{ss}C_{tail}}$$

Substituting Z_{ss}

$$A_{vcm}(s) \approx \frac{-R_D}{2Z_{ss} + 1/g_{m1}}$$

$$A_{vcm}(s) \approx \frac{-R_D}{2 \left(\frac{R_{ss}}{1+sR_{ss}C_{tail}} \right) + \frac{1}{g_{m1}}}$$

$$A_{vcm}(s) \approx \left(\frac{-R_D}{2R_{ss} + 1/g_{m1}} \right) \left[\frac{1 + sR_{ss}C_{tail}}{1 + s \frac{R_{ss}C_{tail}}{1+2g_{m1}R_{ss}}} \right]$$

$$\omega_z = \frac{1}{R_{ss}C_{tail}}$$

$$\omega_p = \frac{1 + 2g_{m1}R_{ss}}{R_{ss}C_{tail}} \approx \frac{2g_{m1}}{C_{tail}}$$

- **Input Pair Transconductance (g_{m1}):** From XM1/XM4, $g_{m1} = 285.2$
- **Tail Output Resistance (R_{ss}):** From the tail current source XM2, $g_{ds2} = 3.437$
- **Load Resistance (R_D):** $R_D = 30k$
- **Tail Capacitance (C_{tail}):** Summing the drain capacitance of the tail and the source capacitances of the input pair:

$$C_{tail} = c_{dd2} + c_{ss1} + c_{ss4}$$

$$C_{tail} = 6.471 \text{ fF} + 20.98 \text{ fF} + 20.98 \text{ fF} = 48.431 \text{ fF}$$

B. DC Common-Mode Gain ($A_{vcm}(0)$)

Using the factored DC component from the transfer function:

$$A_{vcm}(0) = \frac{-R_D}{2R_{ss} + 1/g_{m1}}$$

$$A_{vcm}(0) = \frac{-30 \times 10^3}{2(290.95 \times 10^3) + (1/285.2 \times 10^{-6})}$$

$$A_{vcm}(0) = \frac{-30 \times 10^3}{581.9 \times 10^3 + 3.506 \times 10^3}$$

$$A_{vcm}(0) \approx -0.0512 \text{ V/V} \quad (\approx -25.8 \text{ dB})$$

C. Common-Mode Zero (f_z)

$$\omega_z = \frac{1}{R_{ss}C_{tail}}$$

$$\omega_z = \frac{1}{(290.95 \times 10^3)(48.431 \times 10^{-15})} \approx 70.96 \times 10^6 \text{ rad/s}$$

$$f_z = \frac{\omega_z}{2\pi} \approx 11.29 \text{ MHz}$$

D. Common-Mode Pole (f_p)

First, calculate the $2g_{m1}R_{ss}$ loop gain factor:

$$2g_{m1}R_{ss} = 2(285.2 \times 10^{-6})(290.95 \times 10^3) \approx 165.96$$

Now, calculate the pole frequency:

$$\omega_p = \frac{1}{R_D C_L}$$

$$\omega_p = \frac{1}{(30 \times 10^3)(1 \times 10^{-12})} \approx 11.85 \times 10^9 \text{ rad/s}$$

$$f_p = \frac{\omega_p}{2\pi} \approx 1.88 \text{ GHz}$$

E. Output Pole (f_p)

$$\omega_p = \frac{1 + 2g_{m1}R_{ss}}{R_{ss}C_{tail}}$$

$$\omega_p = \frac{1 + 165.96}{(290.95 \times 10^3)(48.431 \times 10^{-15})} \approx 33.33 \times 10^6 \text{ rad/s}$$

$$f_p = \frac{\omega_p}{2\pi} \approx 5.4 \text{ MHz}$$

Quantity	Hand Analysis	Simulation
A_{vcm} (dB)	-25.8	-28.3
CMRR (dB)	$17.93 - (-25.8) = 43.7$	46.18

Table 7: CM gain & CMRR: hand analysis vs simulation

Is $A_{vcm} < 1$? Why: Yes, $|A_{vcm}| \approx 0.038 \text{ V/V}$, .because R_{ss} is large = 291 k.

Justify the variation of A_{vcm} vs frequency: at dc the tail capacitance is open circuit only RSS exist so the gain degrades, at high frequency the tail cap become short circuit so the gain increase and the CMRR decreases. $|A_{vcm}|$ rises with frequency since the zero comes first then the $R_D \parallel C_L$ output pole takes over and rolls everything back down.

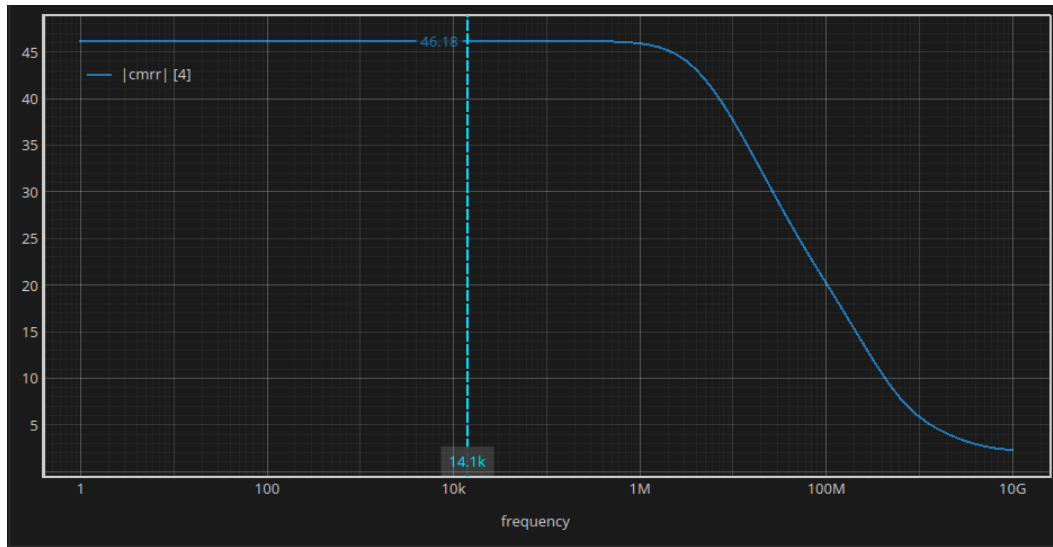


Figure 9: A_{vd}/A_{vcm} (CMRR) in dB vs frequency

Justify the variation of CMRR vs frequency $\text{CMRR} = A_{vd}(\text{dB}) - A_{vcm}(\text{dB})$. A_{vd} stays flat until its own pole at 5.68 MHz then rolls off, while A_{vcm} is rising since there is a zero. at high freq the A_{vd} reduces so CMRR reduces also.

2.5 Differential Large-Signal Characteristics

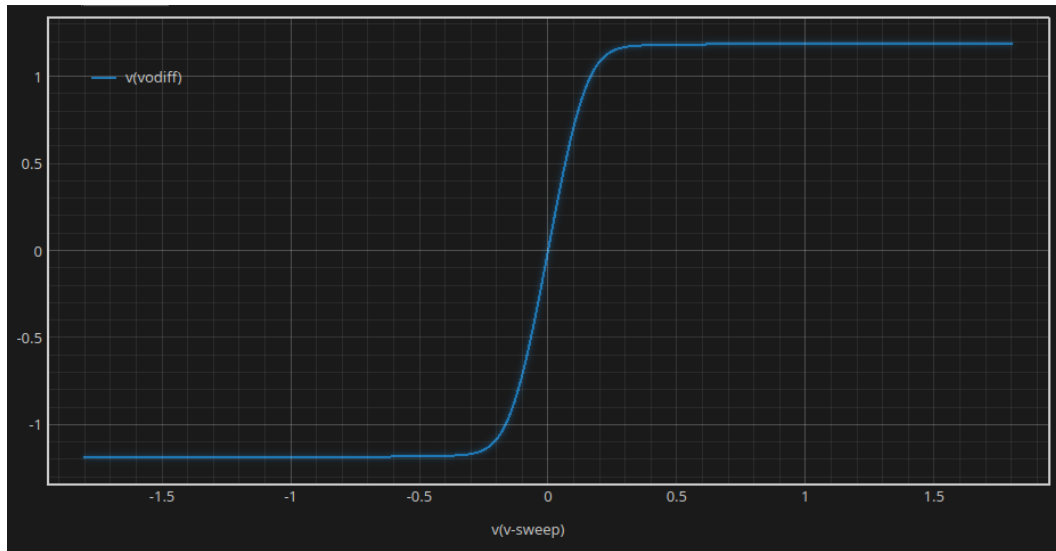


Figure 10: Large-signal differential transfer characteristic (VODIFF vs VIDIFF)

A: The output is $\pm 1.2\text{V}$. when V_{id} is negative then M4 is off and M1 is on so the $V_{odiff} = -I_{SS} \times 30\text{k} = -1.2$ and when V_{id} is positive then M1 is off and M4 is on so the $V_{odiff} = I_{SS} \times 30\text{k} = 1.2$

$$\Delta V_{OUT,max} = I_{SS} \times R_D = 40\mu\text{A} \times 30\text{k} = 1.2\text{ V} \quad (8)$$



Figure 11: deriv(VODIFF) vs VIDIFF, peak = 7.8 V

what's the peak value? Why?

A: The peak is 7.8, because the derivative is the gain so it equals 7.8 at $V_{id}=0$

2.6 CM Large-Signal Characteristics – Region vs V_{ICM}

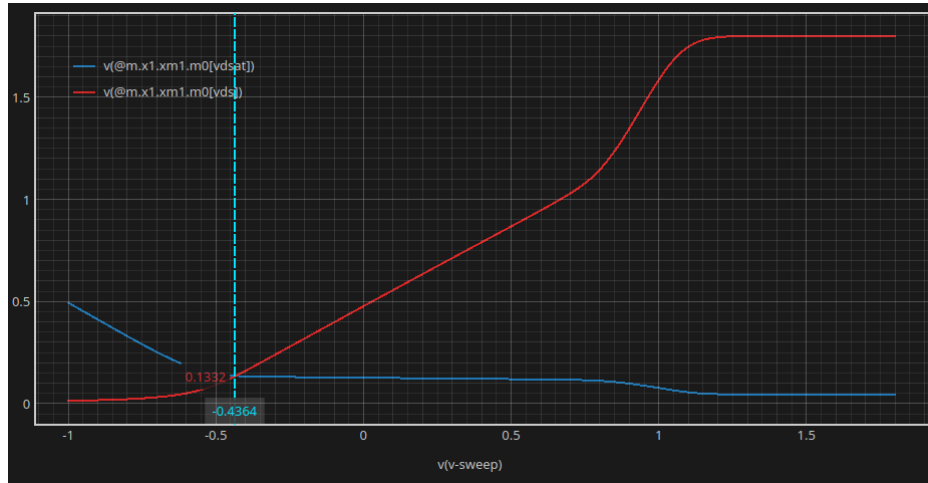


Figure 12: Input pair (M1): V_{DS} vs V_{DSAT} vs V_{ICM} – crossing at $V_{ICM} = -0.436\text{V}$ sets $V_{ICM,min}$

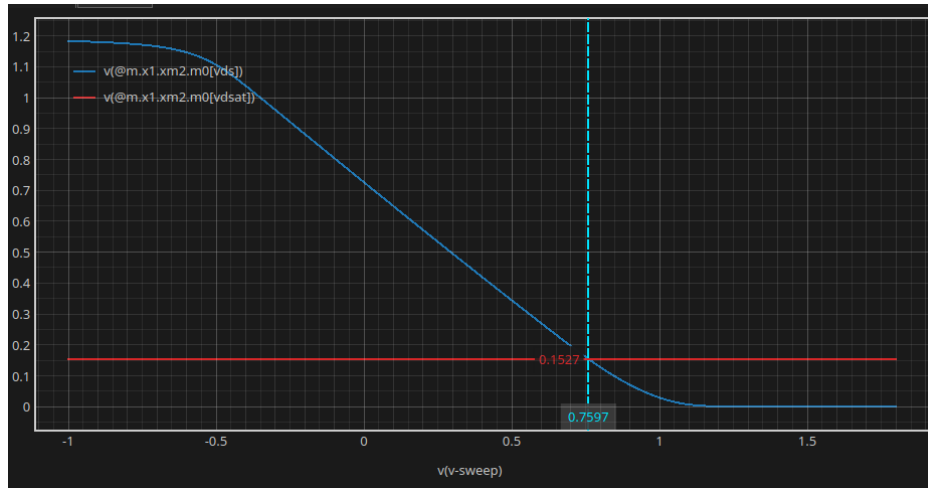


Figure 13: Tail transistor (M2): V_{DS} vs V_{DSAT} vs V_{ICM} – crossing at $V_{ICM} = 0.760\text{V}$ sets $V_{ICM,max}$

A: simulation: $V_{ICM,min} = -0.436\text{V}$ (input pair leaves saturation) and $V_{ICM,max} = 0.760\text{V}$ (tail transistor leaves saturation) $\Rightarrow$ CMIR = 1.196V wide.

Why the min VICM is not what I expected: because we must see the CMIR which GBW is flat not just by taking the vds when its bigger than vdsat.

Method	$V_{ICM,min}$	$V_{ICM,max}$	CMIR width
V_{DS} vs V_{DSAT}	-0.436 V	0.760 V	1.196 V
hand analysis	-0.235 V	0.706 V	0.941 V

Table 8: CMIR comparison: device-saturation method vs hand

2.7 CM Large-Signal Characteristics – GBW vs V_{ICM}

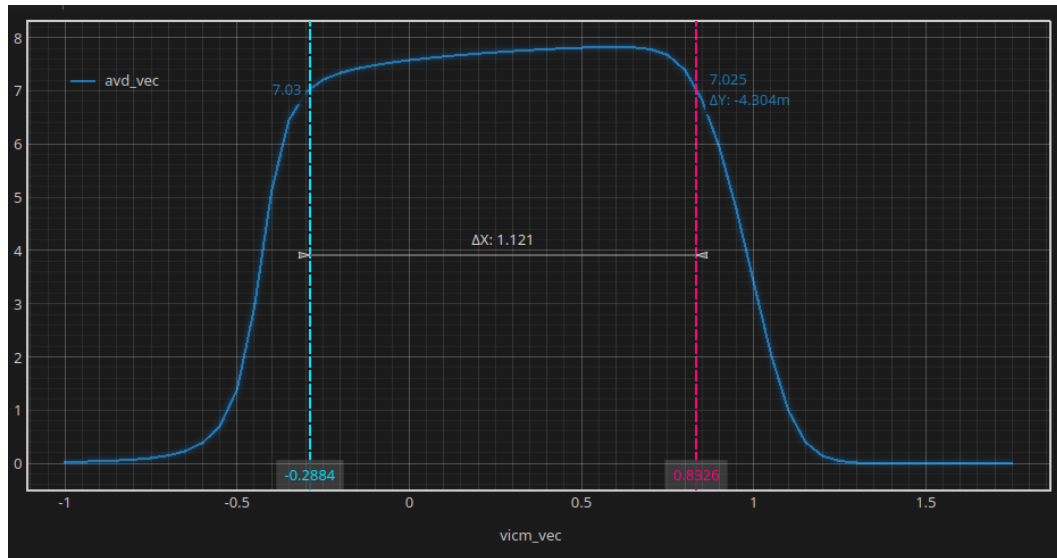


Figure 14: A_{vd} vs V_{ICM} – cursors mark the 10% gain-drop points that define CMIR

A: $V_{ICM,min} = -0.288\text{V}$ and $V_{ICM,max} = 0.833\text{V}$, CMIR width $\Delta X = 1.121\text{V}$.

Method	$V_{ICM,min}$	$V_{ICM,max}$	CMIR width
V_{DS} vs V_{DSAT}	-0.436 V	0.760 V	1.196 V
A_{vd} 10% drop	-0.288 V	0.833 V	1.121 V

Table 9: CMIR comparison: device-saturation method vs functional-gain method

3 Bonus Part: Wide-Swing Cascode Tail Current Source

replacing the simple 2-transistor tail mirror with a full wide-swing cascode: a bottom mirror pair (XM3 = reference, XM2 = mirror output, $2\times$ the width), a cascode pair stacked on top of them (XM6, XM5), and a “magic-battery” diode-connected transistor (XM7) that generates the bias voltage V_B feeding both cascode gates, an NMOS (XM0, XM8, XM10) current mirror we made in lab 05 for biasing the diode connected and the wide swing circuit.

Sizing g_m/I_D and the 0.5V compliance split

$$V_{star} = 20\text{mV} \implies g_m/I_D = 10 \text{ V}^{-1} \quad (9)$$

same as the main lab.

$$V_{DS2} = V_{star} + 50\text{mV} = 0.25\text{V} \implies V_{DS,cascode} = 0.5 - 0.25 = 0.25\text{V} \quad (10)$$

Sizing the magic battery M5 (XM7)

XM7 is same width as the tail’s unit device (WP1 = $41.28\mu\text{m}$ from the main lab) and swept L ($I_D=20\mu\text{A}$, $V_{SB}=0$, $V_{DS}=1.3\text{V}$ since

$$V_{GS} = V_{DS2} + V_{GS6} = 1.05 + 0.25 = 1.3 : \quad (11)$$

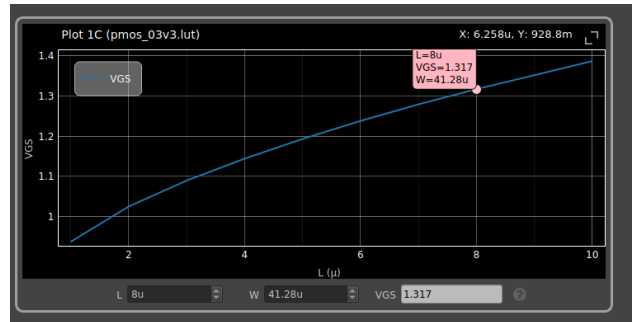
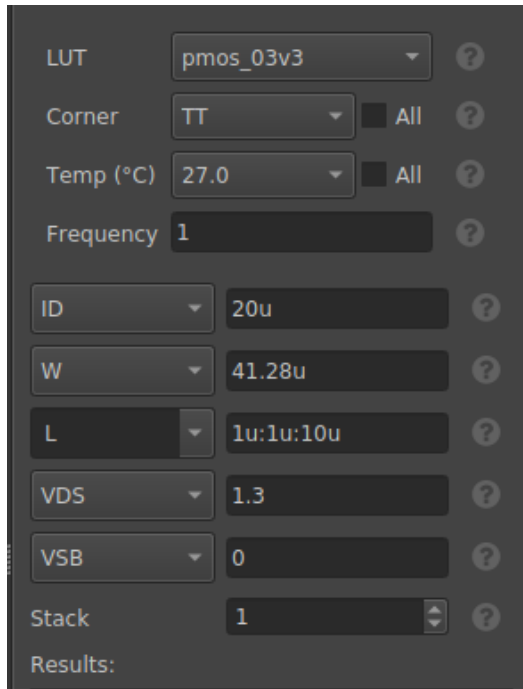


Figure 15: SA sweep of V_{GS} vs L for the magic battery (M5/XM7), $W=WP1=41.28\mu\text{m}$ fixed so $L=8\text{u}$

New CM input level

input-pair $V_{GS} = 941.1\text{mV}$ from the main lab with the new 0.5V tail compliance:

$$V_{ICM} = 1.8 - 0.5 - 0.9411 = \boxed{0.359 \text{ V}} \quad (12)$$

3.1 Operating Point

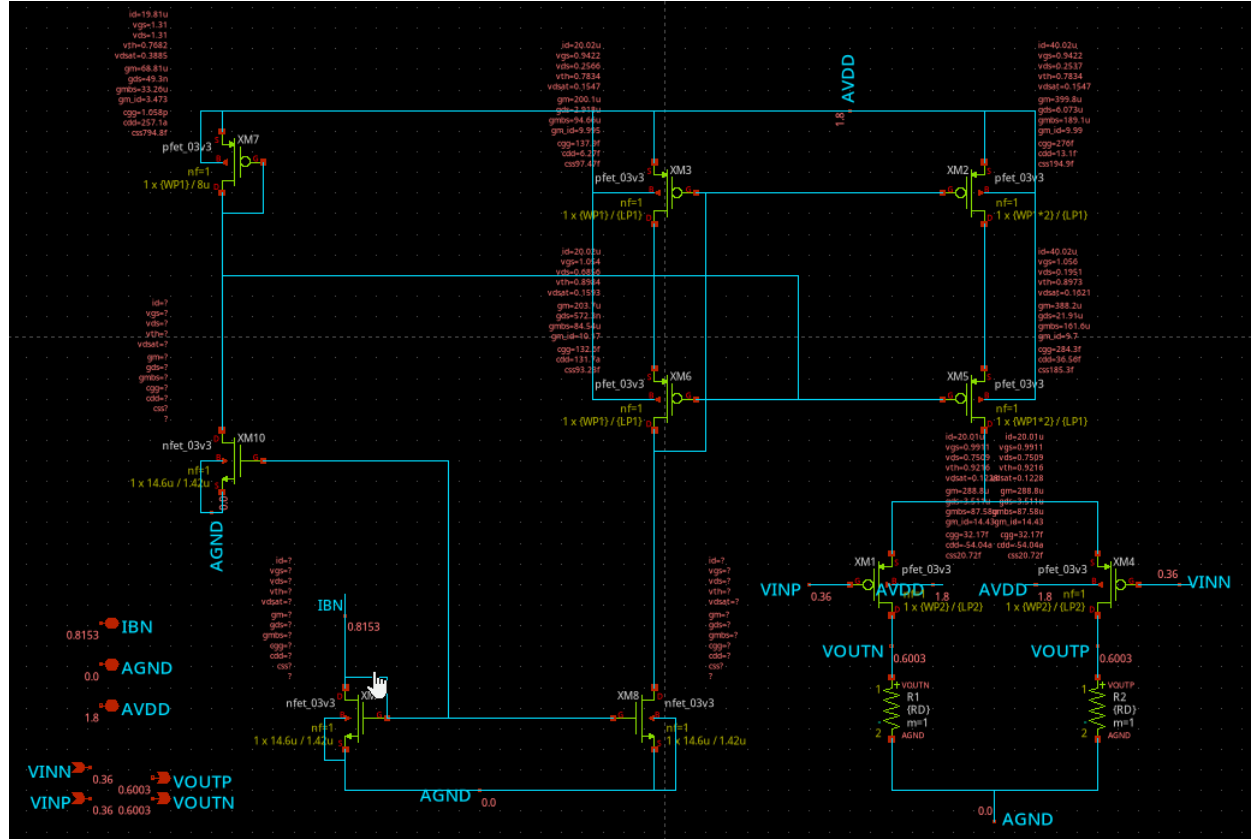


Figure 16: Annotated OP point of the wide-swing cascode tail (XM7=M5 battery, XM3/XM2=bottom mirror, XM6/XM5=cascode pair)

Param	XM7 (M5, battery)	XM3 (bottom, ref)	XM2 (bottom, out)	XM6 (cascode, ref)
I_D	19.81 μA	20.02 μA	40.02 μA	20.02 μA
V_{GS}	1.310 V	940.9 mV	942.2 mV	1.064 V
V_{DS}	1.310 V (diode)	256.6 mV	253.7 mV	654.6 mV
V_{TH}	768.2 mV	783.4 mV	783.4 mV	898.4 mV
V_{DSAT}	388.5 mV	154.7 mV	154.7 mV	191.3 mV
g_m/I_D	3.47	≈ 10.0	≈ 9.99	≈ 10.05
G_M	68.81 $\mu\text{A/V}$	200.1 $\mu\text{A/V}$	399.8 $\mu\text{A/V}$	201.2 $\mu\text{A/V}$
G_{DS}	49.3 nA/V	$\approx 3.9 \mu\text{A/V}$	8.073 $\mu\text{A/V}$	572.3 nA/V
G_{MB}	33.28 $\mu\text{A/V}$	94.6 $\mu\text{A/V}$	189.1 $\mu\text{A/V}$	84.53 $\mu\text{A/V}$

Table 10: OP values

A: Saturation check: all four wide-swing devices show $V_{DS} > V_{DSAT}$ so its saturated. XM7 is diode-connected. **Body effect check:** XM6 shows $V_{TH} = 898.4\text{mV}$, higher than XM3/XM2's 783.4mV since the sources float above ground while their bulk stays tied to V_{DD} .

3.2 A_{vcm} vs Frequency

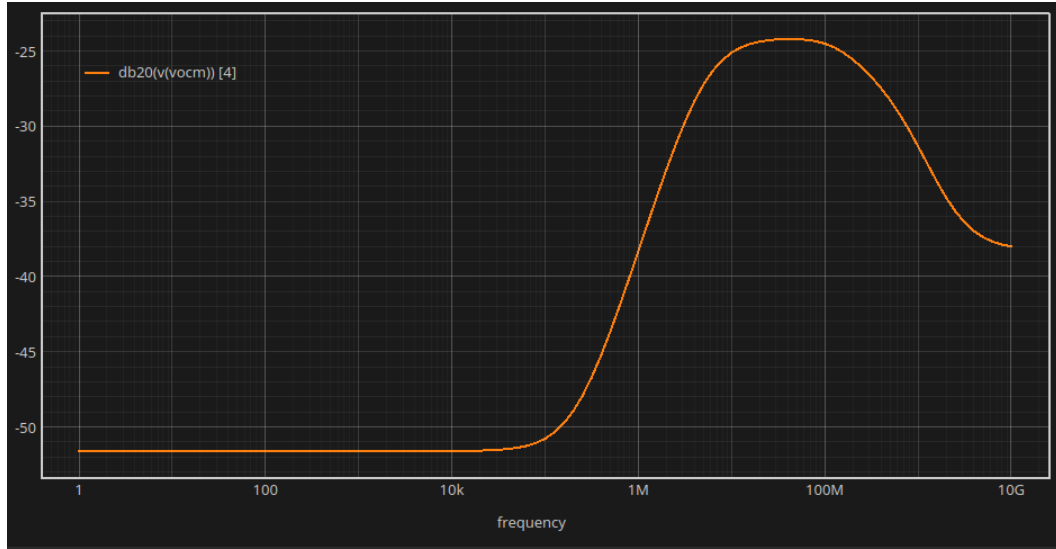


Figure 17: CM AC gain of the diff amp with the wide-swing cascode tail

The DC is -51.5 dB,

cascode boosts the tail's output impedance from just $r_{o,mirror}$ to $g_{m,cascode} \cdot r_{o,cascode} \cdot r_{o,mirror}$. (XM2: $r_{o,mirror} = 1/G_{DS,2} = 1/8.073\mu = 123.9\text{k}\Omega$; XM6: $g_{m,cascode} = 201.2\mu$, $r_{o,cascode} = 1/572.3n = 1.748\text{M}\Omega$):

$$R_{ss,new} \approx g_{m6} r_{o6} r_{o2} = 201.2\mu \times 1.748\text{M} \times 123.9\text{k} \approx 43.6 \text{ M}\Omega \quad (13)$$

$$A_{vcm,hand} \approx \frac{-R_D}{2R_{ss,new}} = \frac{-30\text{k}}{2(43.6\text{M})} = -3.44 \times 10^{-4} \text{ V/V} = -69.3 \text{ dB} \quad (14)$$

Design	$A_{vcm}(\text{DC})$	CMRR
Simple mirror tail	-28.3 dB	46.18 dB
Wide-swing cascode tail (bonus)	-51.5 dB	$\approx 69.4 \text{ dB}$

Table 11: CMRR improvement from adding the wide-swing cascode tail

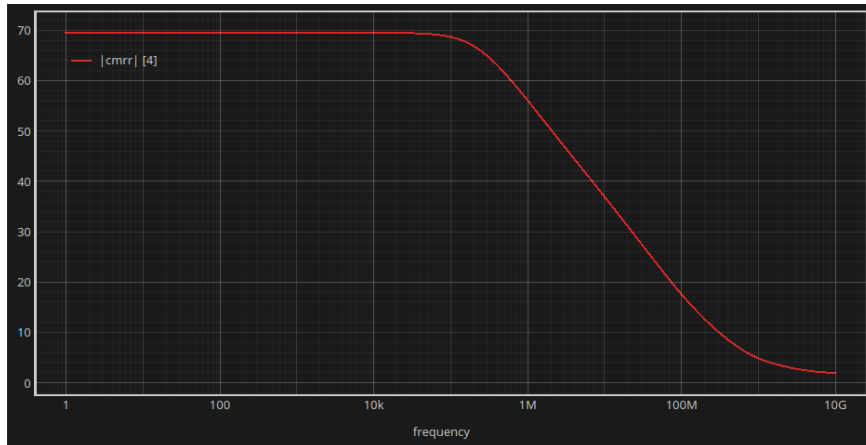


Figure 18: CMRR